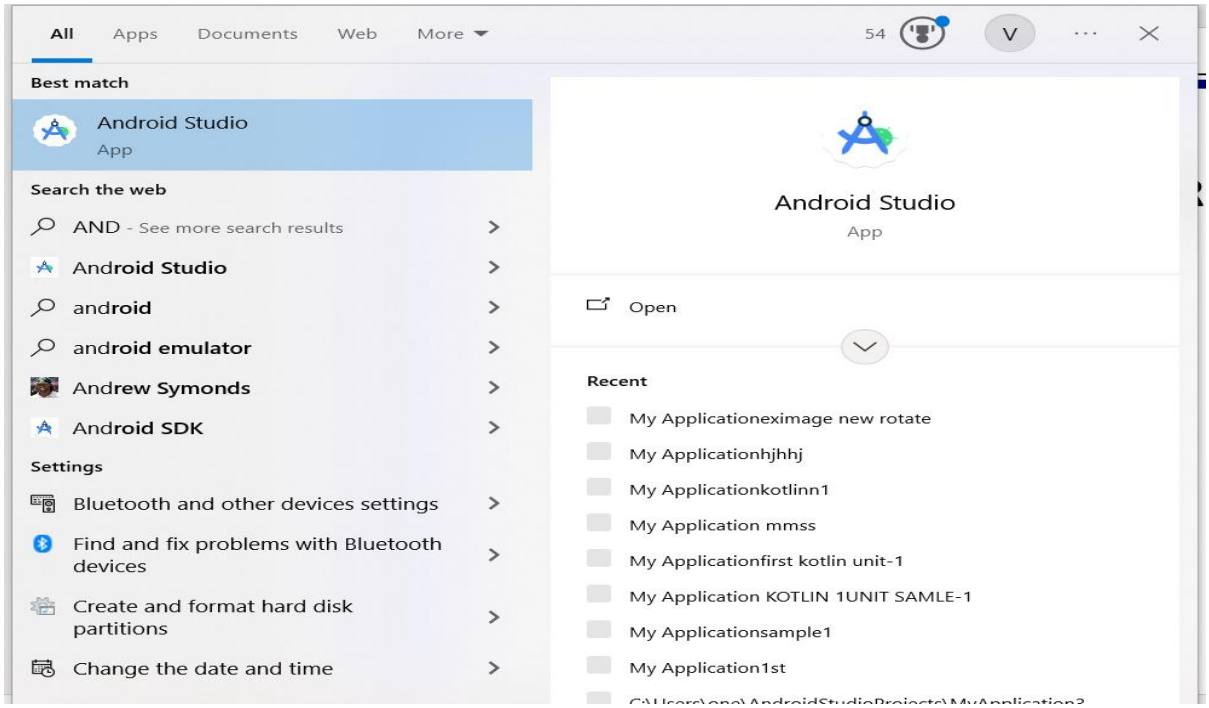
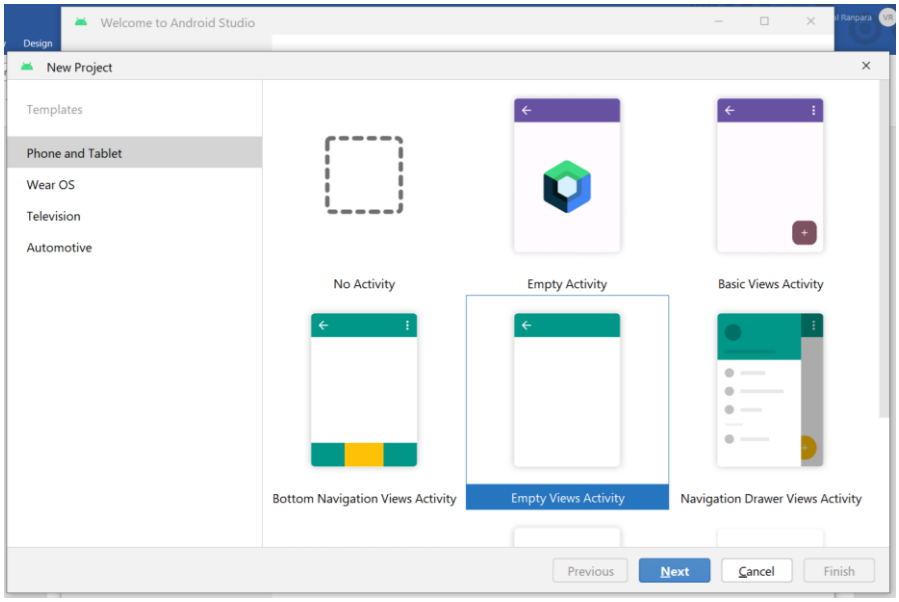
■ HOW TO OPEN ANDROID STUDIO?

STEP-1:-

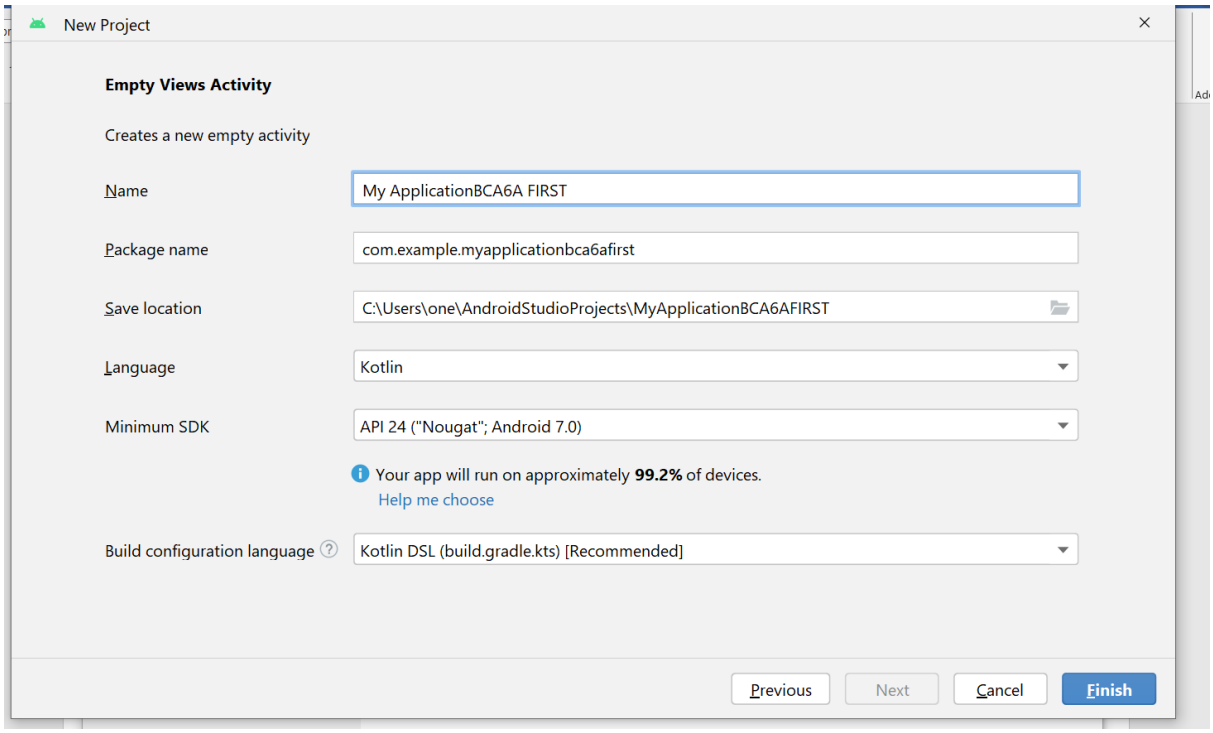


STEP-2:-

SELECT NEW PROJECT & CLICK NEXT & SELECT EMPTY VIEWS ACTIVITY & CLICK NEXT.



STEP:3



➔ **Then write code and design.**

----important Code understanding-------(just for Reference)

- **findViewById:-**
  - ➔ **findViewById is a method used to connect XML UI elements with Kotlin/Java code in Android.**
  - ➔ **All UI controls are written in XML, but logic is written in Kotlin.**
  - ➔ **findViewById acts like a bridge between them.**

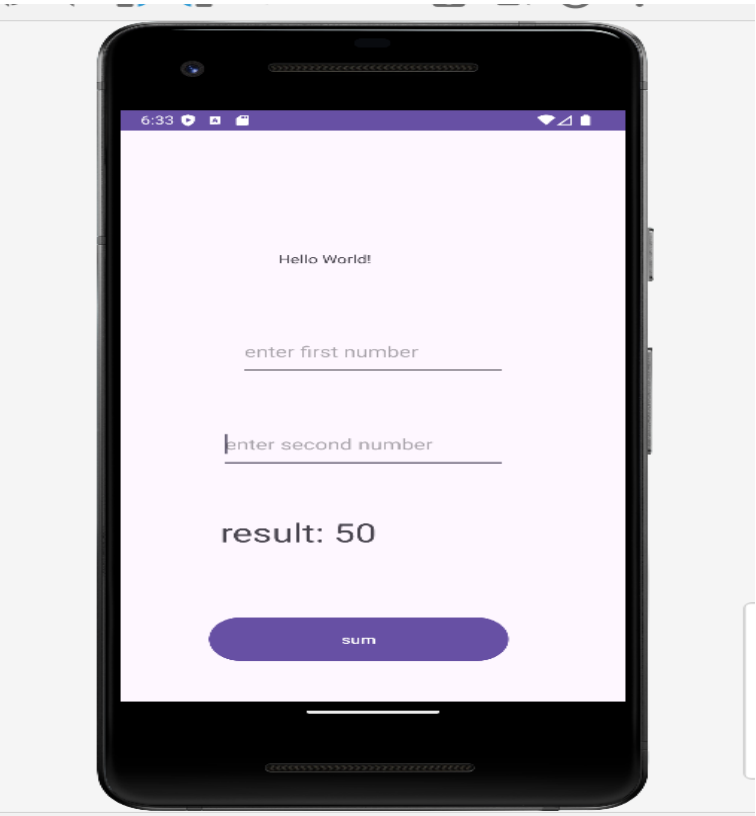
- **findViewById<EditText>(R.id.*t1*)**
  - **R is an auto-generated class in Android.**
  - **Android automatically creates it to store references to all resources in your project.**

- R**.id** is a **subcategory inside the R class** that contains **all view IDs** defined in XML.
  - So: **R.id.t1 means**
    "the ID named t1 that is defined in my XML layout"

- **Toast.makeText(this, "You pressed the button", Toast.LENGTH_SHORT).show():-**
  - ➔ **This line shows a small popup message on the screen for a short time.**
  - ➔ **Toast.LENGTH_SHORT**
    Message display duration
    LENGTH_SHORT → ~2 seconds
    LENGTH_LONG → ~3.5 seconds
  - ➔ **This:- means current activity current page.**

- **Practical Write in Journal:-**

1) **Addition of two number**



- **Xml file**

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/androi
d"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```

```
    android:layout_marginTop="144dp"
    android:text="Hello World!"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.383"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent" />

<TextView
    android:id="@+id/e1"
    android:layout_width="229dp"
    android:layout_height="0dp"
    android:layout_marginStart="81dp"
    android:layout_marginBottom="59dp"
    android:text="result"
    android:textSize="31dp"
    app:layout_constraintBottom_toTopOf="@+id/bt"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/t2" />

<EditText
    android:id="@+id/t1"
    android:layout_width="217dp"
    android:layout_height="0dp"
    android:layout_marginTop="235dp"
    android:layout_marginBottom="51dp"
    android:ems="10"
    android:hint="enter first number"
    android:inputType="number"
    android:text=""
    app:layout_constraintBottom_toTopOf="@+id/t2"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent" />
```

```
<EditText
    android:id="@+id/t2"
    android:layout_width="0dp"
    android:layout_height="0dp"
    android:layout_marginBottom="54dp"
    android:ems="10"
    android:hint="enter second number"
    android:inputType="number"
    android:text=""
    app:layout_constraintBottom_toTopOf="@+id/e1"
    app:layout_constraintEnd_toEndOf="@+id/t1"
    app:layout_constraintStart_toStartOf="@+id/e1"
    app:layout_constraintTop_toBottomOf="@+id/t1" />

<Button
    android:id="@+id/bt"
    android:layout_width="244dp"
    android:layout_height="0dp"
    android:layout_marginStart="72dp"
    android:layout_marginBottom="45dp"
    android:text="sum"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/e1" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

■ Kotlin File:-

```
package com.example.myapplicationsumoftwonumber

import androidx.appcompat.app.AppCompatActivity
```

```
import android.os.Bundle
import android.widget.Button
import android.widget.EditText
import android.widget.TextView

class MainActivity : AppCompatActivity() {
   override fun onCreate(savedInstanceState: Bundle?) {
      super.onCreate(savedInstanceState)
      setContentView(R.layout.activity_main)
      val a = findViewById<EditText>(R.id.t1)
      val b = findViewById<EditText>(R.id.t2)
      val c = findViewById<Button>(R.id.bt)
      val d = findViewById<TextView>(R.id.e1)
      c.setOnClickListener {
         val e = a.text.toString().toInt()
         val f = b.text.toString().toInt()
         val sum = e + f
         d.text = "result: $sum"
      }

   }
   }
```

2. **when you press button then give message "you pressed button"**

Xml:

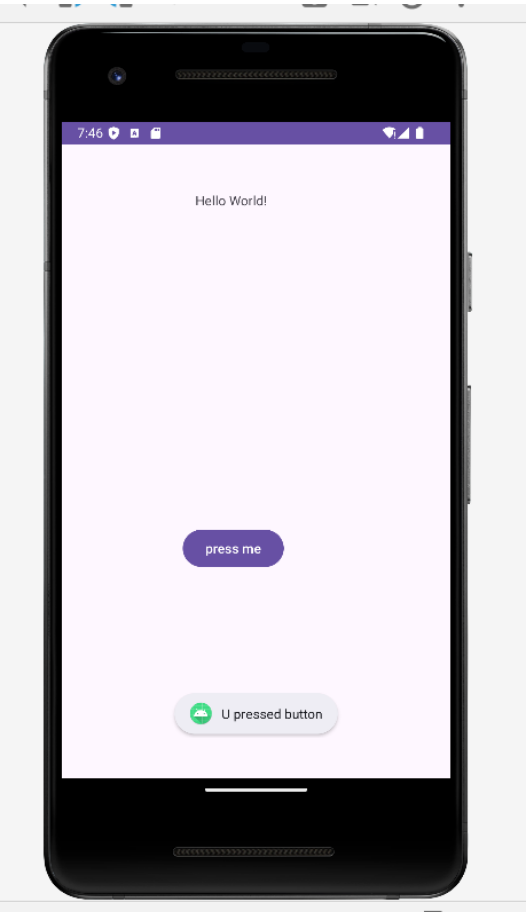
```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   android:layout_width="match_parent"
   android:layout_height="match_parent">

   <Button
      android:id="@+id/bt"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Press Me"

      app:layout_constraintTop_toTopOf="parent"
      app:layout_constraintBottom_toBottomOf="parent"
      app:layout_constraintStart_toStartOf="parent"
```

```
        app:layout_constraintEnd_toEndOf="parent"/>

</androidx.constraintlayout.widget.ConstraintLayout>
```

Kotlin file:-

```
package com.example.myapp

import android.os.Bundle
import android.widget.Button
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val a = findViewById<Button>(R.id.bt)

        a.setOnClickListener {
            Toast.makeText(this, "You pressed the button",
Toast.LENGTH_SHORT).show()
```

```
        }
    }
}
```


7:46
Hello World!
press me
U pressed button

3. When you click a button, the background color of the screen changes.

Xml file:-

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
   android:id="@+id/ly "
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:gravity="center"
   android:orientation="vertical"
   android:background="#FFFFFF">
```

```
    <Button
        android:id="@+id/bt"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Change Background Color" />
</LinearLayout>
```

Kotlin file:-

```
package com.example.backgroundcolorchange

import android.graphics.Color
import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
import android.widget.Button
import android.widget.LinearLayout

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
```

```
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val a = findViewById<LinearLayout>(R.id.ly)
        val b = findViewById<Button>(R.id.bt)

        b.setOnClickListener {
            a.setBackgroundColor(Color.CYAN)
        }
    }
}
```